PARTIAL SHEET ONLY. X1 is an explicit off-sheet electrical contract, NOT implemented gating circuitry.

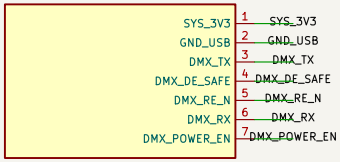
X1 maps to power-service X_DMX: POR/cutoff gate DE and isolated-power enable; resident loader may enable RDM.

U5 pin8 ENABLE INPUT via R4=1k (TI SLLSF86D p5 permits series limiting). Brownout operation unverified.

GND_USB and GND_ISO must remain separate. ISO_RETURN_REFERENCE declares converter return only.

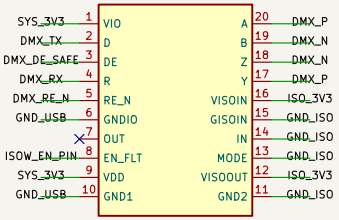
Termination: two240R in parallel via manual S_TERM; assemble OFF. No external TVS; on-chip ESD basis only.

X1



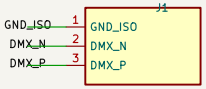
OFF-SHEET CONTROL / POWER CONTRACT

U5



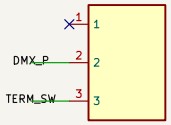
ISOW1412DFMR

J1



XLR1/2/3 boundary

S_TERM



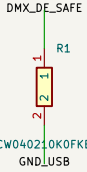
PCM12SMTR OFF=1-2 / ON=2-3

#FLG01



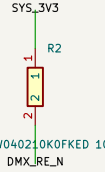
ISOLATED RETURN REFERENCE
GND_ISO

DMX_DE_SAFE



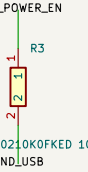
CRCW040210K0FKED 10k 1%
GND_USB

SYS_3V3



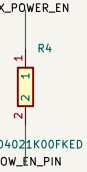
CRCW040210K0FKED 10k 1%
DMX_RE_N

DMX_POWER_EN



CRCW040210K0FKED 10k 1%
GND_USB

DMX_POWER_EN



CRCW04021K00FKED 1k 1%
ISOW_EN_PIN

SYS_3V3



GRM155R71E104KE14D 100nF 25V X7R
GND_USB

SYS_3V3



GRM21BR71A106KE51L 10uF 10V X7R
GND_USB

ISO_3V3



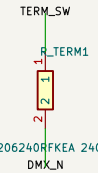
GRM155R71E104KE14D 100nF 25V X7R
GND_ISO

ISO_3V3



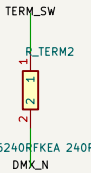
GRM21BR71A106KE51L 10uF 10V X7R
GND_ISO

TERM_SW



CRCW1206240RFKEA 240R 1% 0.25W
DMX_N

TERM_SW



CRCW1206240RFKEA 240R 1% 0.25W
DMX_N

Sheet: /
File: dmx-interface.kicad_sch

Title: NOVA partial isolated DMX interface - NOT RELEASED

Size: A4	Date:	**Rev: C-TERM1**
KiCad E.D.A. 10.0.7		Id: 1/1